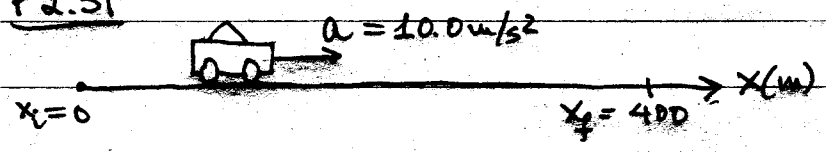
Solutions HW04 (1) S&F; ch.2
Starykh

P2.31



Choose $x_i = 0$, then $x_f - x_i = 400\,m$; initial velocity $v_i = 0\,m/s$.

(b) Known displacement $x_f - x_i$ and acceleration a

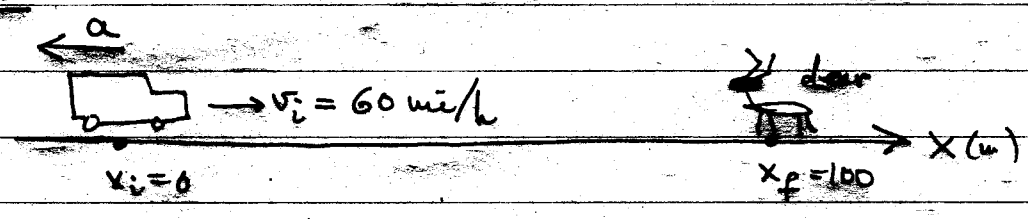
$$\Rightarrow \quad v_f^2 = v_i^2 + 2a(x_f - x_i)$$

$$\Rightarrow \quad v_f^2 = 2 \cdot 10\,m/s^2 \cdot 400\,m = 8000\,m^2/s^2$$

$$\Rightarrow \quad \boxed{v_f = 89.4\,m/s}$$

(a) To find time: $v_f = \overset{=0}{v_i} + a(t_f - t_i) \Rightarrow t_f - t_i = \frac{v_f}{a} = \frac{89.4\,m/s}{10\,m/s^2} = 8.94\,s$

P2.33



In order to stop the car has to have $v_f = 0\,m/s$ after traveling $x_f - x_i = 100\,m$. Initial velocity $v_i = 60\,mi/h =$

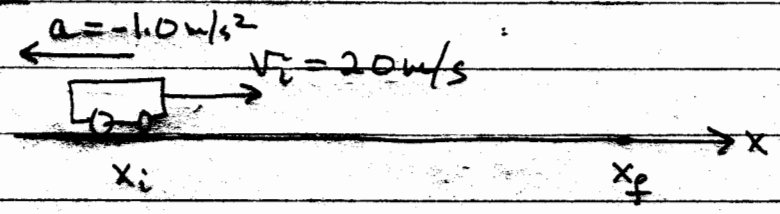
$= 60 \cdot \frac{1609\,m}{3600\,s} = 26.8\,m/s$. Acceleration a has to be <u>negative</u>

$$\Rightarrow \quad 0 = v_f^2 = v_i^2 + 2a(x_f - x_i)$$

$$\Rightarrow \quad a = -\frac{v_i^2}{2(x_f - x_i)} = -\frac{(26.8\,m/s)^2}{2 \cdot 100\,m} = \boxed{-3.6\,m/s^2}$$

$a < 0$ because velocity is decreasing for the slowing car.

P2.35



Given: at $t_i = 0\,s$ $v_i = 20\,m/s$, $a = -1.0\,m/s^2$

Find: displacement $x_f - x_i$ during $t_f - t_i = 40\,s$ from the moment the brakes are applied.

Let's find how long it will take the train to stop:

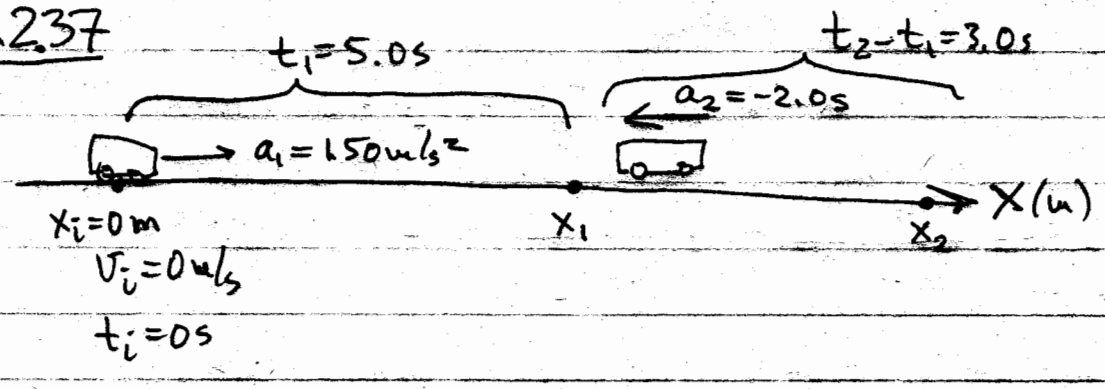
$$0 = v_f = v_i + at \Rightarrow t = -\frac{v_i}{a} = -\frac{20\,m/s}{-1.0\,m/s^2} = 20\,s$$

$\Rightarrow$ train is moving only $t = 20\,s$ from $t_i = 0$!

Its displacement is $x_f - x_i = v_i t + \frac{1}{2} a t^2 = 20 \cdot 20 - 0.5 \cdot (20)^2 =$
$= 400 - 200 = 200\,m$.

Thus, $\boxed{x_f - x_i = 200\,m}$ because train moves only 20s, not 40s as the problem suggests.

P.237



Denote X_1 distance traveled during first $t_1 = 5.0\,s$ ($a_1 = +1.50\,m/s^2$) and X_2 distance traveled during next $t_2 - t_1 = 3.0\,s$ ($a_2 = -2.0\,m/s^2$)

Find velocity v_1 at $t_1 = 5.0\,s$: $v_1 = \overset{=0}{v_i} + a_1 t_1 = 1.50 \cdot 5 = 7.50\ m/s$

position x_1 at t_1: $x_1 = \overset{=0}{v_i} t_1 + \frac{1}{2} a_1 t_1^2 = 18.75\ m$

For the time interval $t_2 - t_1 = 3.0\,s$ v_1 = initial velocity

x_1 = initial position

$\Rightarrow$ velocity v_2 at the end of the second interval

$v_2 = v_1 + a_2 (t_2 - t_1) = 7.50 - 2.0 \cdot 3.0 = 1.50\ m/s$

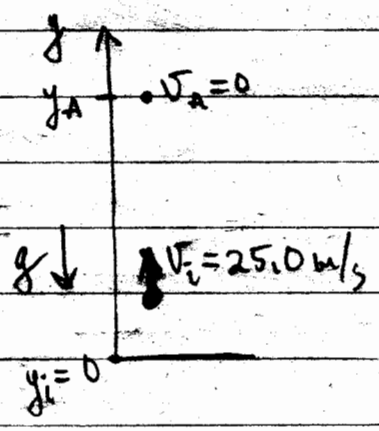
position $x_2 = x_1 + v_1 (t_2 - t_1) + \frac{1}{2} a_2 (t_2 - t_1)^2 =$

$= 18.75 + 7.50(3.0) - \frac{2.0}{2}(3.0)^2 = 18.75 + 13.5 = 32.25$

$\Rightarrow$ total distance traveled $\boxed{x_2 - x_i = 32\ m}$

velocity at $t_2 = 8.0\,s$ $\boxed{v_2 = 1.50\ m/s}$

P 2.43



Free fall acceleration $g = -9.80\ m/s^2$

Denote A the highest point — at that point velocity $v_A = 0\ m/s$.

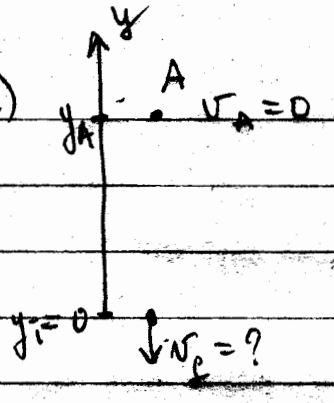
(a) $0 = v_A^2 = v_i^2 + 2g\, y_A \Rightarrow y_A = -\frac{v_i^2}{2g} = -\frac{25^2}{-2 \cdot 9.80} =$

$\Rightarrow \boxed{y_A = 31.9\ m}$

(b) How long to reach y_A? $0 = v_A = v_i + g t_A \Rightarrow t_A = -\frac{v_i}{g} = -\frac{25.0}{-9.80} =$

$= \boxed{2.55\,s}$

(c)



Time from A to the ground?

$$y_i = y_A + \underset{0}{v_A} t + \frac{1}{2} g t^2$$

$$\Rightarrow y_i - y_A = \frac{1}{2} g t^2$$

$$\Rightarrow t^2 = \frac{2(y_i - y_A)}{g} = \frac{2(0-31.9)}{-9.80} = 6.51$$

$$\Rightarrow t = \sqrt{6.51} = \boxed{2.55\,s}$$

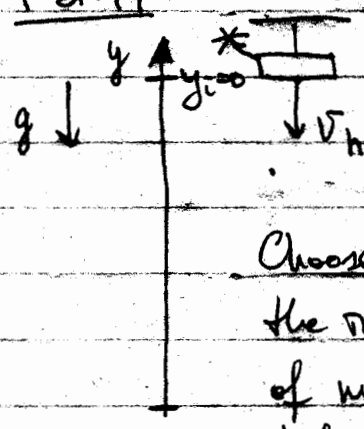
(d) Velocity v_f at $y_i = 0$?

$$v_f = v_A + g t = -9.80 \cdot 2.55 = \boxed{-25.0\ m/s}$$

Note: time to go up ($t_A = 2.55s$) and down ($t = 2.55s$) are equal!

Final velocity $v_f = -25.0\ m/s$ differs in sign only from initial $v_i = 25.0\ m/s$: the minus sign simply tells us that the motion is in negative y direction.

P 2.47



Choose $y_i = 0$ the initial position of mailbag and helicopter.

(a) Mailbag is droped from the helicopter moving down, hence its initial velocity $v_{bag,i} = -1.50\ m/s$ (the same as helicopter's)

After $t = 2.0\,s$ $v_{bag,f} = v_{bag,i} + g t =$
$= -1.50 - 9.80 \cdot 2.0 = -21.1\ m/s$

$\boxed{v_{bag,f} = -21.1\ m/s}$ The speed $= |v_{bag,f}| = 21.1\ m/s$

(b) How far below is it from the helicopter?

Helicopter's displacement during $t = 2.0\,s$ is

$$y_{hel,f} - \overset{=y_i}{y_{hel,i}} = + v_{hel} \cdot 2 = -1.50 \cdot 2 = -3.00\,m$$

Mail bag's displacement during the same time is

$$y_{bag,f} - \overset{=y_i}{y_{bag,i}} = + v_{bag,i} \cdot t + \frac{1}{2} g t^2 = -1.50 \cdot 2 - \frac{9.80}{2}(2)^2 =$$

$$= -3.00 - 19.6 = -22.6\,m$$

Distance between the helicopter and the mail bag is

$$d = (y_{bag,f} - y_i) - (y_{hel,f} - y_i) =$$

$$= -22.6 - (-3.00) = -19.6\,m$$

$\Rightarrow$ the mail bag is $\boxed{19.6\,m}$ below the helicopter

(c) If the helicopter is rising instead, then its velocity has <u>positive</u> sign $v_{hel} = 1.50\,m/s$

$\Rightarrow$ $v_{bag,i} = +1.50\,m/s$ also.

$\Rightarrow$ $v_{bag,f} = v_{bag,i} + gt = 1.50 - 9.80 \cdot 2 = -18.1\,m/s$

$\Rightarrow$ $\boxed{\text{the speed} = 18.1\,m/s}$

$y_{hel,f} - y_i = +3.00\,m$

$y_{bag,f} - y_i = +1.50 \cdot 2 - \frac{9.80}{2}(2)^2 = 3.00 - 19.6 = -16.6\,m$

The distance $= |-16.6 - (+3.00)| = \boxed{+19.6\,m}$ remains the same!